DAY 14 · THEORY HANDOUT

How a Transformer Generates Text

Follow one text path, compare model types, and connect the parts to BERT, GPT, and small language models.

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1. Encoder and decoder roles

An encoder makes contextual input representations. A decoder builds an output sequence.

Encoder: reads

BERT-style model.

Uses context on both sides.

Makes text representations.

Decoder: continues

GPT-style model.

Uses available earlier text.

Predicts the next token.

Encoder + decoder

One reads the source.

One produces new text.

Useful for translation.

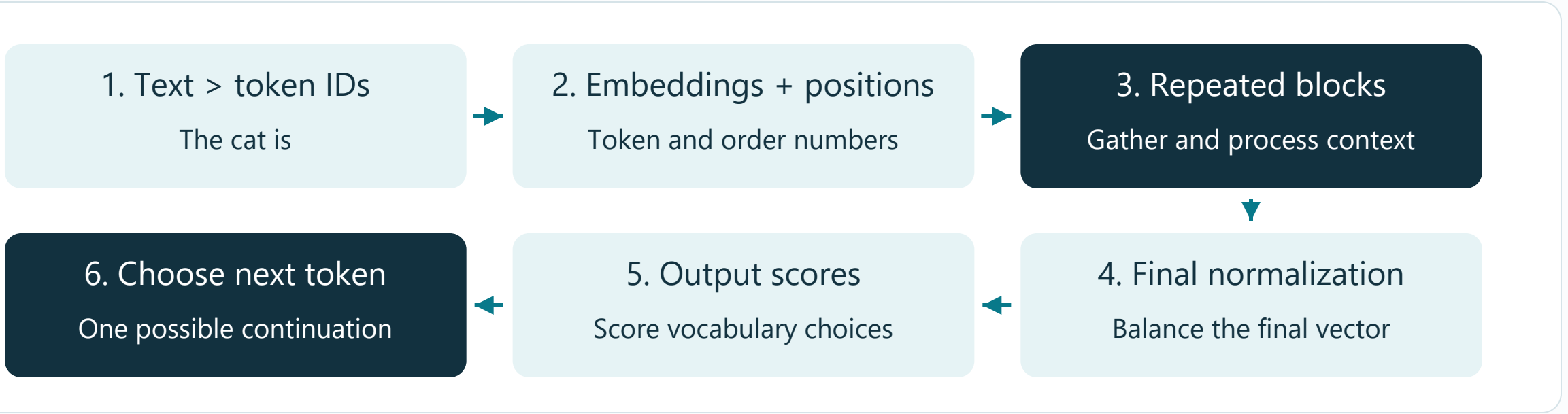
BERT means Bidirectional Encoder Representations from Transformers. "Bidirectional" means using left and right context. It can support text labels, search, and finding answer spans.

GPT means Generative Pre-trained Transformer. It is a decoder-style model. In an encoder-decoder model, cross-attention lets the decoder use encoder information.

> A decoder model is more than converting token IDs back to letters. It learns to predict output tokens.

2. The full GPT-style path

A prompt is the input text. Every stage changes it into numbers used for a continuation.



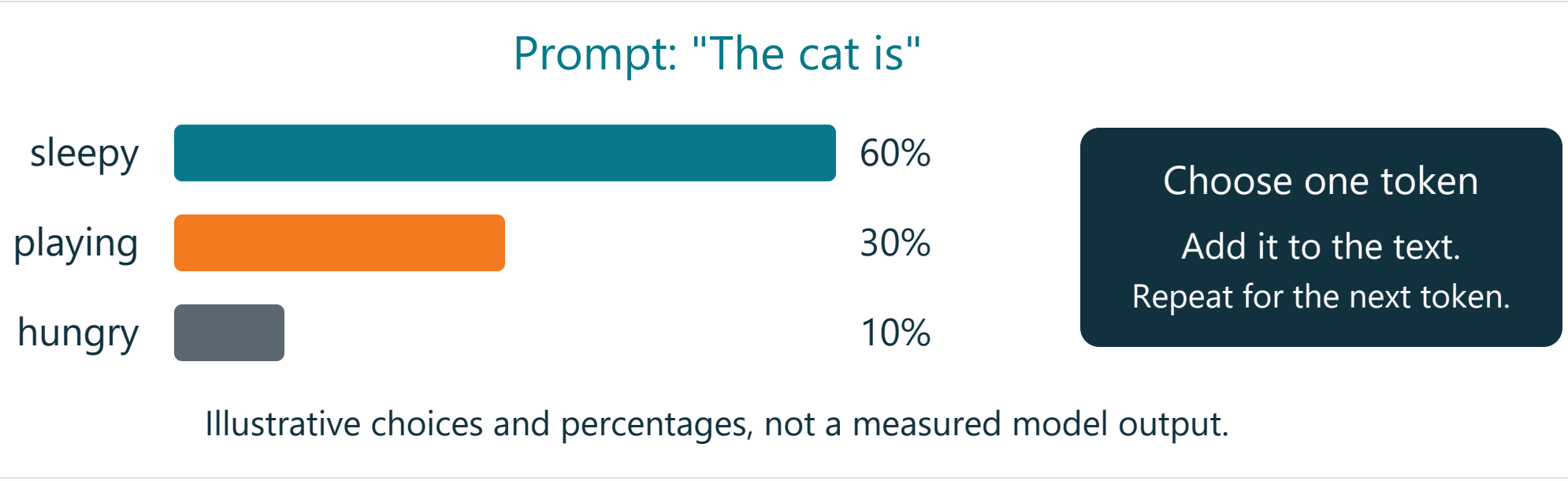
Inside each block: attention mixes available context, feed-forward processing changes each vector, and shortcuts preserve earlier inputs.

The output layer scores possible next tokens. The tokenizer maps the chosen ID back to a text piece. Detokenization joins the pieces into text.

> This is the generation path for a GPT-style model. An encoder-only model uses its representations for a different output task.

3. Scores, probabilities, and generation

Logits are raw output scores. Softmax converts the scores into probabilities that add up to 100%.



Illustrative choices and percentages, not a measured model output.

Sampling chooses a token using the probabilities. Some settings take the top choice; others allow less likely choices. The highest score need not always win.

Temperature changes how spread out the choices are. Lower values favor the leading choices more; higher values allow more variety. It does not make answers more truthful.

> The selected token becomes part of the context for the next step. Generation ends at a stop token or a length limit.

4. Learning and using the model

Pre-training builds a starting model. Fine-tuning trains it further. Inference means using it.

Pre-training

Learn patterns from text.
Weights change during
many training examples.

Fine-tuning

Train further on selected
examples or feedback.
Adapt behavior to a task.

Inference

Use trained weights.
Process our prompt.
Produce a model output.

GPT-style pre-training learns next-token patterns. BERT-style pre-training includes predicting hidden text pieces using surrounding context.

During inference, our prompt changes the context and internal representations. Ordinary generation does not update the trained weights. Fine-tuning does.

> A language model does not automatically search the web. A separate tool can supply information for it to use.

5. A short 2017 to 2022 history

BERT and the first GPT appeared before 2022. ChatGPT's 2022 launch brought chat-style use to many people.

2017	Transformer	An attention-based encoder-decoder design.
2018	BERT and GPT	BERT: both-side context. GPT: text continuation.
2019	GPT-2	A later GPT model; small versions help visual demos.
2022	ChatGPT	Launched November 30, based on a GPT-3.5-series model.

BERT is an encoder-style model family. GPT is a decoder-style family. ChatGPT is a chat product using further-trained models, not another name for BERT.

> This is a short historical snapshot. The dates do not describe every model or the current ChatGPT system.

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6. Small and large language models

A parameter is a learned number in a model, such as a weight in an embedding or attention layer.

SLM

Small Language Model.
Fewer learned parameters.
Often needs less memory.

LLM

Large Language Model.
More learned parameters.
Often needs more memory.

Same core idea

Both can use Transformers.
Size does not alone decide which task it does well.

SLM and LLM describe relative size, not two separate attention methods. Data, training, hardware, and the task also matter.

Transformer Explainer uses GPT-2 small, with about 124 million parameters. It is an older teaching model that makes the processing steps visible.

> A small model can make fluent mistakes too. Check its answers using the task and examples that matter.

7. Explain the whole Transformer

Follow the path by naming the job of each part. Keep token preparation separate from model processing.

Tokenizer	Split text and assign IDs.
Embedding + position	Give tokens numeric input and order.
Attention + heads	Mix available context through learned views.
Feed-forward / MLP	Process each token representation.
Shortcuts + normalization	Preserve earlier inputs and balance scale.
Output layer + softmax	Score tokens and form probabilities.
Sampling + repeat	Choose a token, append it, then continue.

For GPT-style generation, the causal mask blocks later positions. The current output depends on the available prompt and earlier generated tokens.

For BERT-style tasks, both-side context helps build representations. An encoder-decoder model can read one sequence and produce another.

> Useful language patterns can produce useful answers. They do not guarantee facts, human understanding, or reliable reasoning.

DAY 14 · COMPLETE

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